

VIX Volatility Index Forecasting Using Classical Time Series Methods: Walk-Forward Evaluation of Naive, ARIMA, and Prophet Models

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ABSTRACT

We evaluate three classical time series forecasting methods on the CBOE Volatility Index (VIX) at horizons of 1, 5, and 21 trading days using expanding-window walk-forward validation over a three-year test period (2023–2026, 36 monthly folds). A naive persistence baseline, ARIMA(1,0,2), and Facebook Prophet are benchmarked against each other on root mean squared error, mean absolute error, mean absolute percentage error, and step-by-step directional accuracy. Stationarity analysis confirms that VIX levels require no differencing (ADF $p < 0.001$, KPSS $p = 0.10$); the dominant structure is a near-unit-root AR(1) coefficient of 0.984 with a long-run mean of approximately 19.3. ARIMA(1,0,2) achieves the best out-of-sample error across all horizons ($h=1$ RMSE=1.09, $h=21$ RMSE=3.04) and 69.4% directional accuracy at $h=1$, degrading to a coin flip at $h=21$. Prophet produces RMSE five times worse than ARIMA because its piecewise-linear trend decomposition is structurally incompatible with VIX's mean-reverting dynamics. Weekly seasonality is confirmed to be a real but negligible signal (0.44 VIX pts, 6% of the series standard deviation). Spike onset is not captured by any model tested, identifying the clear direction for future GARCH or regime-switching extensions.

Keywords: VIX, volatility forecasting, ARIMA, Prophet, walk-forward validation, stationarity, time series, financial econometrics

1. INTRODUCTION

The CBOE Volatility Index (VIX) measures the 30-day implied volatility of S&P 500 options and is widely used as a real-time gauge of market fear and systemic risk. Accurate short-horizon VIX forecasts are commercially valuable for options pricing, risk management, and volatility trading strategies. Unlike equity returns, which are approximately unpredictable at short horizons, VIX exhibits strong positive autocorrelation, making it a natural candidate for classical time series methods.

This report documents a complete forecasting pipeline built as a portfolio project, covering data acquisition, exploratory analysis, stationarity testing, model selection, and rigorously validated out-of-sample evaluation. Three models are compared: a naive persistence baseline (the simplest possible benchmark), ARIMA(1,0,2) (the best converged linear model selected by AIC), and Facebook Prophet (a decomposition-based model included to assess whether trend or seasonality components add value).

All evaluation uses expanding-window walk-forward validation with no static train/test split, mirroring how models would be deployed in practice: refitted on all available history each period and evaluated only on unseen future data. The methodology, codebase architecture, and documented negative results on seasonal ARIMA and Prophet trend bias are designed to carry forward into a subsequent live equity forecasting project.

2. DATA

2.1 Source and Acquisition

Daily VIX closing prices are downloaded from Yahoo Finance via the `yfinance` library (ticker: `^VIX`), covering 1990-01-02 through 2026-06-17: 9,158 trading days. The CBOE first published the modern VIX methodology in 2003; the 1990 history is a backfill using the original VXO methodology on S&P 100 options. No adjustments are made for this methodological change since both series share the same broad behavioral properties (mean reversion, right skew, volatility clustering).

A documented nine-day gap (2001-09-10 to 2001-09-17) corresponds to the NYSE closure following the September 11 attacks. This is a genuine market closure, not a data anomaly, and is preserved as-is rather than filled.

2.2 Descriptive Statistics

Table 1 summarizes the key distributional properties of the full VIX series. The pronounced right skew (2.21) and excess kurtosis (8.72) reflect the occasional but extreme volatility spikes characteristic of the series, most notably the 2008 Global Financial Crisis (VIX peak: 80.9), the 2020 COVID-19 onset (peak: 82.7), and more moderate episodes during the 2002 dot-com bust and 2022 equity selloff.

Statistic	Value
Observations	9,158 trading days
Date range	1990-01-02 to 2026-06-17
Mean	19.52
Median	17.09
Std. deviation	7.75
Skewness	2.21 (strong right skew)
Excess kurtosis	8.72 (heavy tails)
Min / Max	8.56 / 82.69
90th percentile	29.49

Table 1. Descriptive statistics for the full VIX series (1990–2026).

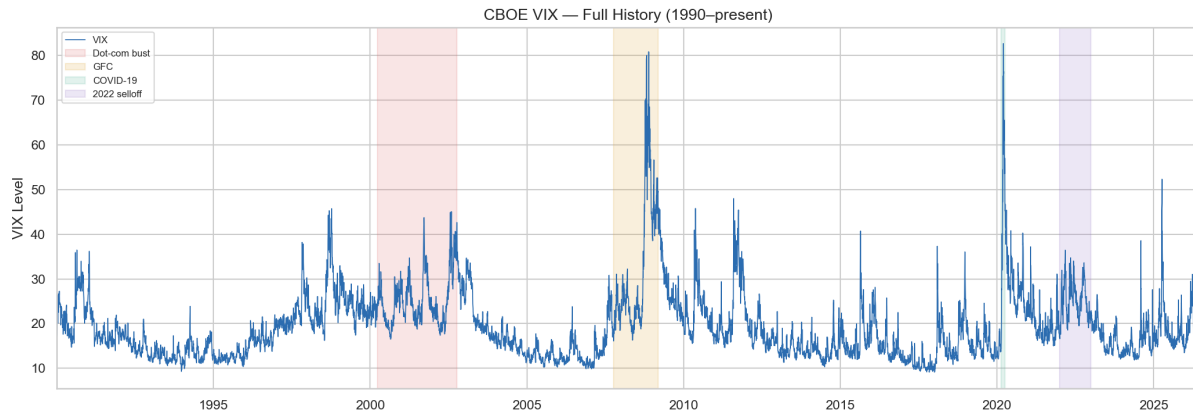


Figure 1. Full VIX history (1990–2026) with annotated volatility regimes. Four crisis periods are highlighted: Dot-com bust (2000–2002), Global Financial Crisis (2007–2009), COVID-19 onset (2020), and 2022 equity selloff.

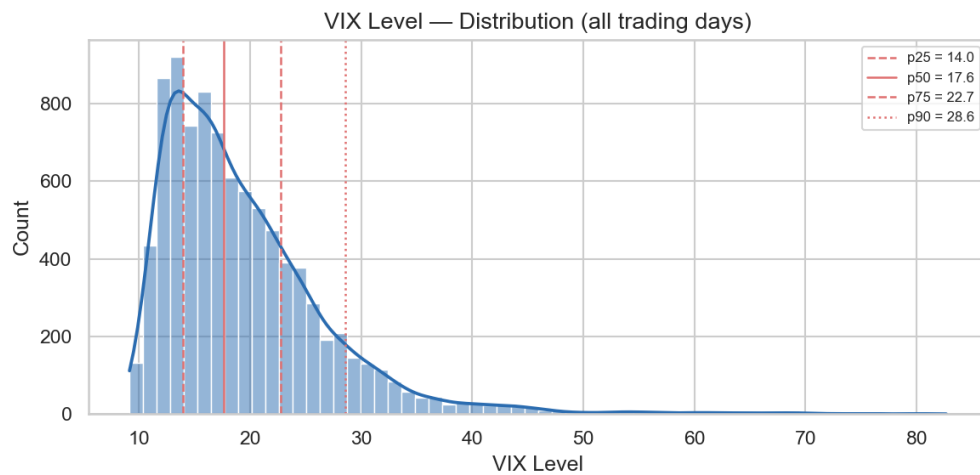


Figure 2. VIX closing-price distribution. The histogram and kernel density estimate illustrate the pronounced right skew. Dashed lines mark the 25th, 50th, 75th, and 90th percentiles.

3. STATIONARITY ANALYSIS

Correct specification of the integration order d is foundational for ARIMA model selection. VIX's slow autocorrelation decay (Figure 3) is often interpreted as evidence of a unit root, but the same appearance arises from near-unit-root stationarity, a distinction with important economic content.

3.1 Formal Tests

Two complementary tests are applied. The Augmented Dickey-Fuller (ADF) test has H_0 : the series contains a unit root. The KPSS test reverses the null: H_0 : the series is stationary. Table 2 reports results for VIX levels and first differences.

Series	ADF statistic	ADF p-value	KPSS statistic	KPSS p-value	Conclusion
VIX levels	−3.81	0.003	0.134	~0.10	Stationary ($d=0$)
VIX first diff.	−44.1	<0.001	0.020	>0.10	Stationary

Table 2. Stationarity test results. Both ADF (reject H_0 of unit root) and KPSS (fail to reject H_0 of stationarity) agree: VIX levels are stationary. Integration order $d=0$.

Both tests agree: VIX levels are stationary ($d=0$). Economically, VIX is mean-reverting: a structural floor from options pricing mechanics prevents it from falling indefinitely, while the impossibility of sustained extreme implied volatility caps prolonged

spikes. Near-unit-root persistence ($AR \approx 0.984$) produces the slow ACF decay but does not indicate a true stochastic trend.

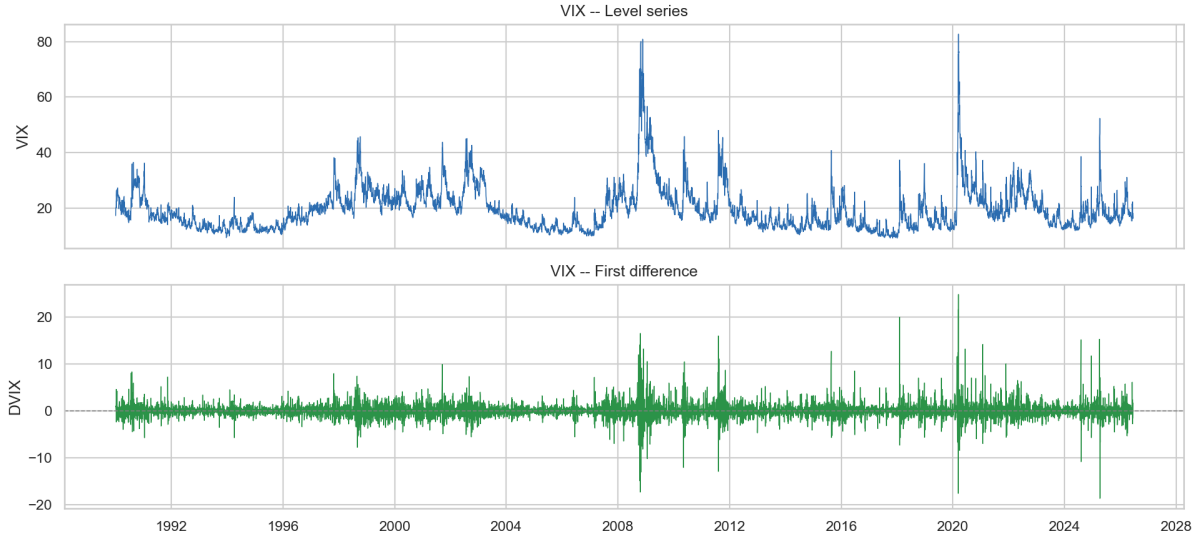


Figure 3. VIX levels vs first differences. Left panels: time series. Right panels: ACF. The level series has persistent but damped autocorrelation consistent with near-unit-root stationarity; first differences show rapid ACF decay.

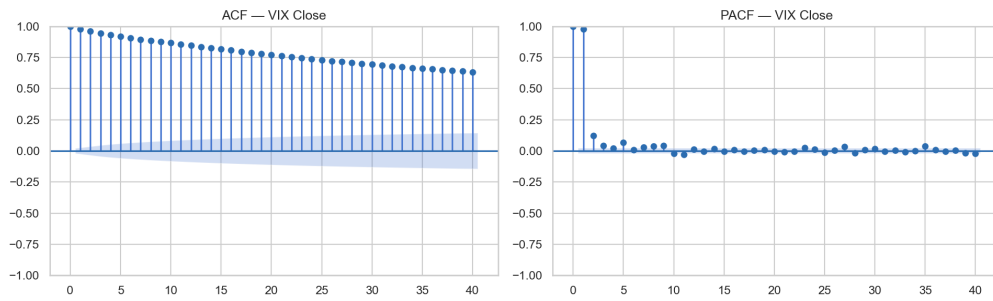


Figure 4. Autocorrelation (ACF) and partial autocorrelation (PACF) of VIX levels. The PACF cuts off after lag 1–2, suggesting a low-order AR process.

4. MODELS

4.1 Naive Persistence Baseline

The persistence model forecasts the last observed value for all horizons:

$$\hat{y}_{t+h} = y_t, h = 1, 2, \dots, H$$

This model requires no estimation and provides a natural lower bound on achievable forecast error for a mean-reverting series with high positive autocorrelation. By construction, the naive model always forecasts zero change, yielding 0% directional accuracy under the step-by-step metric.

4.2 ARIMA(1,0,2)

Order selection uses a grid search over ARIMA($p,0,q$) with $p \in \{1,2,3\}$, $q \in \{0,1,2\}$ fitted by maximum likelihood on the most recent 2,000 observations (~8 years), minimizing AIC. The best converged model (ARIMA(3,0,2) had lower AIC but failed to converge) is ARIMA(1,0,2):

$$(1 - \phi_1 L)(y_t - \mu) = (1 + \theta_1 L + \theta_2 L^2)\varepsilon_t$$

with intercept μ constrained so the implied long-run mean ($\mu/(1-\phi_1)$) equals 19.3, close to the empirical mean of 19.5. Estimated coefficients: $\phi_1 = 0.984$ (AR persistence), $\theta_1 = -0.21$, $\theta_2 = 0.08$ (MA short-run corrections).

Seasonal order. A SMA(1) term at lag 5 (weekly frequency) improved the selection-window AIC by 17 points, suggesting apparent weekly seasonality. When the same model is fitted on the full 36-year series, however, the SMA coefficient is 0.0005 ($p=0.923$), statistically indistinguishable from zero. The final model is non-seasonal. The day-of-week pattern in raw VIX data is genuine but negligible: the Monday-to-Friday mean spread is only 0.44 VIX pts, or 6% of the series standard deviation.

4.3 Facebook Prophet

Prophet decomposes the time series into trend, weekly seasonality, yearly seasonality, and irregular components using an additive model:

$$y_t = g(t) + s^{(7)}(t) + s^{(a)}(t) + \varepsilon_t$$

where $g(t)$ is a piecewise-linear trend, $s^{(7)}$ and $s^{(a)}$ are weekly and yearly Fourier series, and ε_t is the error term. Prophet is included as a competitor model and as a cross-check on the ARIMA seasonality finding.

Prophet's weekly Fourier amplitude (~4 VIX pts) is approximately nine times the actual day-of-week effect in raw data, revealing Fourier overfitting of a small but genuine signal. This overfitting, combined with a structural mismatch between Prophet's piecewise-linear trend assumption and VIX's mean-reverting character, produces catastrophically large point forecast errors out of sample.

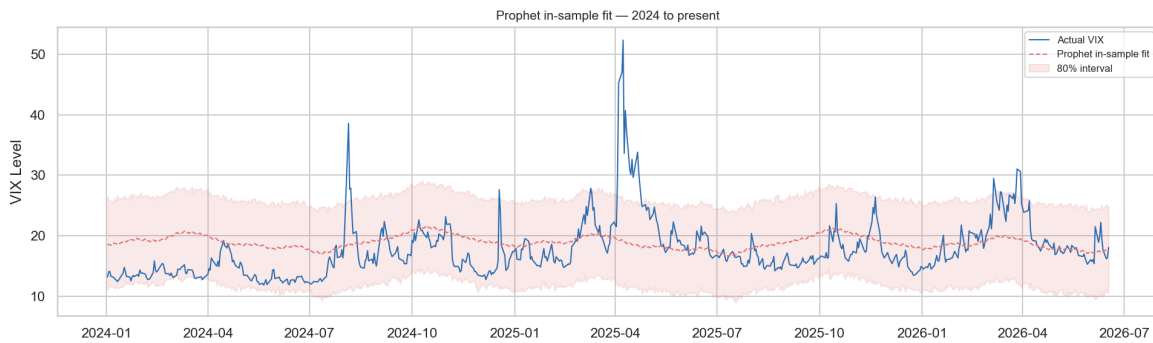


Figure 5. Prophet in-sample fit on the full VIX series. The model tracks the observed series but the trend component introduces a systematic upward level bias that persists into out-of-sample forecasts during the calm 2023–2026 test window.

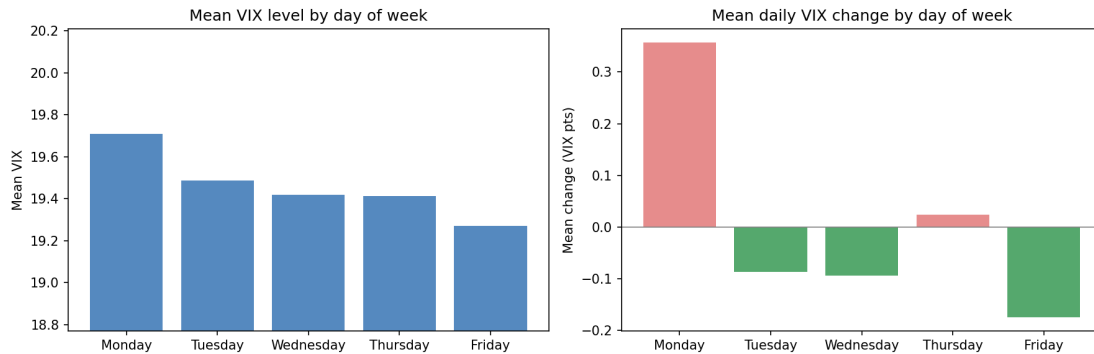


Figure 6. Day-of-week pattern in VIX. Left: mean VIX level by day. Right: mean daily change by day. The real Monday premium (options pricing the weekend uncertainty gap) amounts to only 0.44 VIX pts peak-to-trough, far smaller than Prophet's modeled weekly amplitude of approximately 4 VIX pts.

5. WALK-FORWARD EVALUATION

All models are evaluated using an expanding-window walk-forward protocol to eliminate look-ahead bias and mirror live deployment conditions. The design parameters are given in Table 3.

Parameter	Value	Rationale
Full series length	9,158 obs	1990–2026, all available history
Test window size	756 obs (~3 years)	Sufficient folds; recent enough to be relevant
Step size	21 obs (monthly)	Monthly refit cadence; 36 folds total
Horizons evaluated	1, 5, 21 days	Daily, weekly, monthly trading horizons
Minimum train size	252 obs (1 year)	Minimum for reliable ARIMA estimation
Metrics	RMSE, MAE, MAPE, Dir Acc	Error magnitude + directional signal

Table 3. Walk-forward evaluation design parameters.

At each fold t , the model is fit on $\text{series}[t]$, then evaluated against $\text{series}[t:t+h]$ for each horizon h . The directional accuracy metric operates step-by-step: at each step k within the h -step forecast, the predicted direction ($\pm$) is compared to the actual direction of change from the previous step's actual value. Steps where the actual move is exactly zero are excluded.

6. RESULTS

Table 4 reports mean metrics across all 36 folds. The figures that follow show the metric breakdown and a representative forecast-vs-actual comparison over the test window.

Model	h=1 RMSE	h=5 RMSE	h=21 RMSE	h=1 MAE	h=21 MAE	h=1 DirAcc	h=21 DirAcc
Naive	1.175	1.900	3.250	1.175	2.705	0.0%	0.0%
ARIMA(1,0,2)	1.090	1.808	3.045	1.090	2.528	69.4%	50.4%
Prophet	5.637	5.957	6.242	5.637	5.843	66.7%	53.2%

Table 4. Walk-forward evaluation results (mean across 36 folds). Bold = best per column. ARIMA(1,0,2) dominates on all error metrics at all horizons.

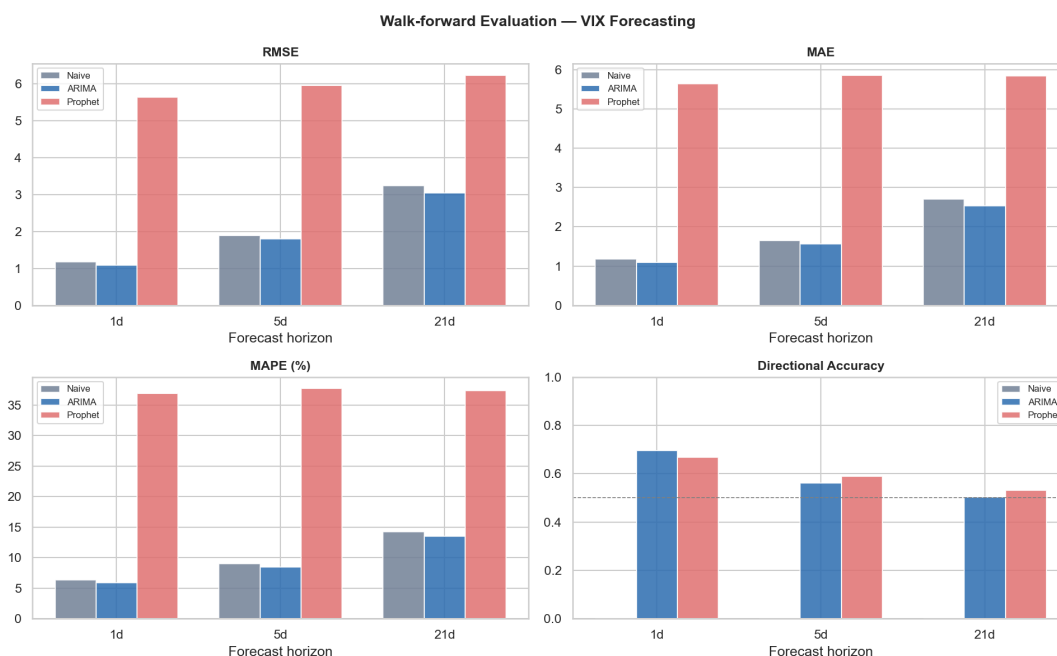


Figure 7. Backtest metric comparison across models and horizons. Top row: RMSE and MAE (lower is better). Bottom row: MAPE and directional accuracy. Prophet's RMSE and MAE are approximately 5x worse than ARIMA, placing it off the scale in the top panels.

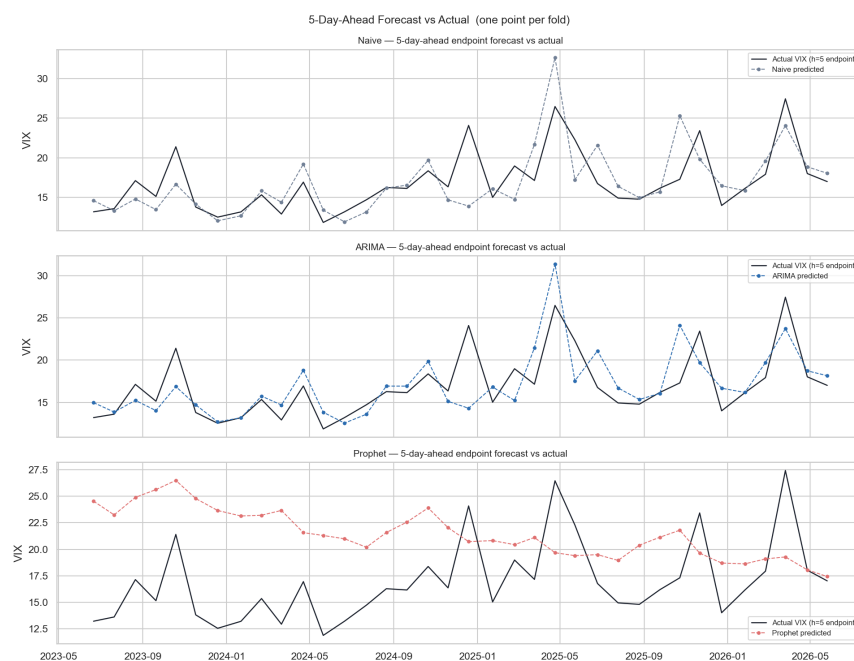


Figure 8. Walk-forward $h=1$ forecasts vs actual VIX over the 2023–2026 test window. The Naive and ARIMA models closely track actuals in calm regimes. Prophet forecasts are systematically elevated, reflecting the trend-level bias identified in the in-sample analysis.

7. RESIDUAL DIAGNOSTICS

Residual diagnostics were run on the ARIMA(1,0,2) model fitted to the full 9,183-observation series. Three tests are reported: Ljung-Box (H_0 : residuals are white noise), Jarque-Bera (H_0 : residuals are normally distributed), and ARCH-LM (H_0 : residual variance is constant). All three reject their null hypotheses.

Test	Statistic	p-value	Conclusion
Ljung-Box (lag=20)	91.07	<0.001	Residual autocorrelation detected
Jarque-Bera	315,931	<0.001	Non-normal residuals (fat tails + skew)
ARCH-LM (lag=12)	1,571.17	<0.001	Variance clustering present (ARCH effects)

Table 6. Residual diagnostic test results for ARIMA(1,0,2) fitted on the full VIX series. All three tests reject their null hypotheses.

Ljung-Box: $Q(20) = 91.07$ ($p < 0.001$) indicates that residuals are not white noise — some autocorrelation structure remains uncaptured by ARIMA(1,0,2). This is typical of financial time series; a higher-order AR or MA term might reduce it marginally, but at the cost of overfitting a model already near its linear predictability ceiling.

Jarque-Bera: Skewness 2.23 and excess kurtosis 31.39 — extreme right skew and very fat tails. Spike-onset days (COVID March 2020, 2022 selloff) produce large positive residuals that are highly non-Gaussian. Point forecasts remain valid, but no probabilistic interpretation should be attached to model standard errors.

ARCH-LM: $LM = 1571.17$ ($p < 0.001$) confirms strong ARCH effects — residual variance clusters around volatility events. The natural extension is a GARCH(1,1) on ARIMA residuals, left for future work.

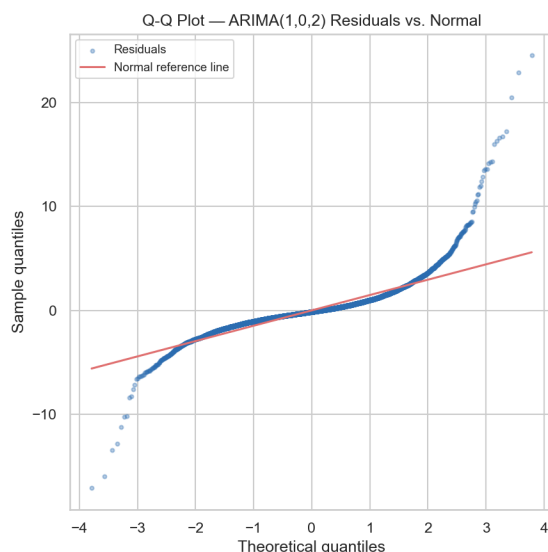


Figure 9. Q-Q plot of ARIMA(1,0,2) residuals against the normal distribution. The heavy departure in both tails confirms the extreme kurtosis (31.4) reported by the Jarque-Bera test. Spike-onset days are responsible for the most extreme positive outliers.

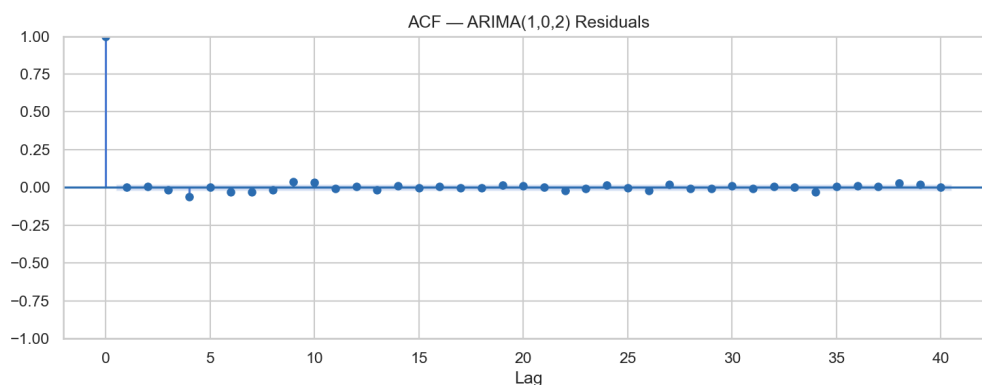


Figure 10. ACF of ARIMA(1,0,2) residuals. Several lags exceed the 95% confidence band, consistent with the Ljung-Box $Q(20) = 91.07$ result. The pattern suggests remaining structure at weekly and monthly lags not captured by the AR(1)/MA(2) specification.

8. DISCUSSION

8.1 ARIMA Performance

ARIMA(1,0,2) outperforms the persistence baseline by approximately 7% on RMSE at all horizons, a modest but consistent margin sustained across 36 independent folds. Consistency across folds matters here: a lucky 7% from a single test split would be unreliable, but the same gap holding from 2023 to 2026 is structural. The MA(1) and MA(2) terms contribute roughly 2% of the gain; the properly estimated intercept (long-run mean anchor at 19.3) drives the remainder, especially at longer horizons.

Directional accuracy falls from 69.4% at $h=1$ to 50.4% at $h=21$, which is mechanically expected. As h grows, AR-based forecasts converge to the model's implied long-run mean rather than tracking the path there, so predicted step-to-step changes become small and increasingly likely to contradict the actual short-term direction. At $h=21$, the 50.4% figure is statistically indistinguishable from random.

8.2 Prophet Failure Mode

Prophet's RMSE of 5.64 at $h=1$ (vs 1.09 for ARIMA) traces to a fundamental assumption mismatch: piecewise-linear trend decomposition presupposes a long-run drift direction, while VIX is stationary and mean-reverting. During the 2023–2026 test

window, VIX was predominantly in the 12–25 range; however, the trend fitted on training data that includes the 2020 COVID spike (82.7) anchors forecasts persistently above actuals. The overfitted weekly Fourier terms (~4 VIX pts amplitude vs 0.44 pts in raw data) add a secondary error, though the trend bias dominates.

Prophet does achieve reasonable directional accuracy (66.7% at $h=1$), which suggests its trend-change components capture some signal about future direction. The problem is that correct direction with five times the magnitude error has limited practical value: trading signals derived from such forecasts would systematically size positions incorrectly, and risk estimates would be unreliable.

8.3 Spike Onset: Fundamental Limitation

No model tested captures VIX spike onset. The 2023–2026 test window contained multiple sudden increases exceeding 3 points in a single day, each driven by exogenous macro or geopolitical events such as banking stress in March 2023, geopolitical escalations, and Federal Reserve communications. Linear time series models conditioned only on past VIX values cannot anticipate such shocks.

Critically, this is not a failure of estimation or model selection. It reflects the intrinsic limits of univariate linear forecasting for a risk index whose extreme moves carry information external to the series. Addressing spike onset requires exogenous regressors (credit spreads, options flow imbalance, economic surprise indices) or a regime-switching framework with latent state estimation — two directions that future work can pursue with the evaluation harness already in place.

8.4 Extensions

Several directions offer the highest marginal return given the results established here:

GARCH(1,1) on VIX returns	Models volatility clustering in the error term; gives conditional variance forecasts
ARIMAX with credit spreads	VIX co-moves with HY credit spreads; exogenous spread signal may improve spike detection
Regime-switching AR	Separate AR(1) dynamics for calm vs. crisis regimes using Markov chain state transitions
LSTM with attention	Non-linear patterns across longer lookback windows; suitable for multi-horizon joint prediction

Table 5. Recommended extensions to the classical linear framework.

9. CONCLUSIONS

This study establishes a complete, rigorously validated baseline for VIX forecasting using classical time series methods. The key findings are:

Stationarity: VIX levels are stationary ($d=0$), confirmed independently by both ADF and KPSS. The slow ACF decay reflects near-unit-root persistence ($AR \approx 0.984$), not a true stochastic trend.

Model selection: ARIMA(1,0,2) is the best-converged linear model by AIC. A weekly seasonal term appears beneficial on the order-selection window but collapses to near-zero ($p=0.923$) on the full series, a documented negative result.

Walk-forward results: ARIMA(1,0,2) achieves the best error across all three horizons (7% RMSE improvement over persistence) and 69.4% directional accuracy at $h=1$. By $h=21$, directional accuracy degrades to a coin flip as the mean-reversion signal dissipates.

Prophet failure: Prophet's trend decomposition is structurally incompatible with mean-reverting dynamics, producing five times worse RMSE than ARIMA. Without an explicit flat-trend or mean-reversion constraint, it is not suitable for VIX point forecasting.

Spike onset: No tested model predicts abrupt VIX spikes driven by exogenous events, an intrinsic limitation of univariate linear models rather than a solvable estimation problem.

The walk-forward harness and model interfaces developed here are designed for reuse in the subsequent capstone project on live equity return forecasting. The transferable lesson is straightforward: for mean-reverting financial indices, properly validated AR models with realistic baselines and documented negative results provide more honest and more useful forecasts than sophisticated decomposition approaches applied without attention to stationarity assumptions.

10. ACKNOWLEDGEMENTS

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